

RRB JE — 25-Day Plan

Science day-wise notes · Maths topics, patterns & formulas · Reasoning patterns

Day 1 = Fri 2 Oct 2026 · Day 25 = Mon 26 Oct · Exam: Tue 27 Oct (confirm on the official RRB site)

How to use this PDF

1. **Science:** read only today's section (about 30 min). After each part, answer the **Check** questions without looking back.
2. **Maths:** this PDF gives the topic, the question patterns and the formulas. Practise from your book, timed.
3. **Reasoning:** this PDF gives the topic and the patterns. Practise from your book, timed.
4. **Current affairs:** 20 MCQs from the monthly PDF every day. Don't read the PDF cover to cover.
5. Log every mistake in the table at the end of this PDF, then redo those questions the next day.

Accuracy note

The CBT-1 structure assumed here (100 Qs / 90 min: Maths 30, Reasoning 25, General Awareness 15, General Science 15, 1/3 negative marking) is from previous RRB JE notifications. **Verify it against your current CEN notice.** Topic order and priorities are my recommendation, not official weightage.

Daily routine (about 4 hours)

#	Block	Time	What you do	Output
1	Maths: 1 topic	75 min	Read the formulas here → 15-20 timed questions from your book → check → log	Score, sec/Q, mistakes
–	Break	10 min	Away from the screen	–
2	Reasoning: 1 topic	50 min	Read the patterns here → 15 timed questions → log	Score, sec/Q, mistakes
–	Break	10 min	Away from the screen	–
3	Science: 1 topic	30-40 min	Today's notes → Check questions → Practice set → answer key	Score, weak facts
4	Current affairs	20 min	20 MCQs from the monthly PDF	Score, retry list
5	Recall	15 min	Redo yesterday's + today's wrong questions with no notes	3 rules for tomorrow

Speed targets: Maths 60-70 sec per question · Reasoning 35-45 sec · Science and GA 15-20 sec.

25-day master plan

Day	Date	Maths	Reasoning	Science
1	Fri 2 Oct	Percentage	Number series	Physics 1: Units, measurement & motion
2	Sat 3 Oct	Ratio & proportion, partnership	Alphabet & alpha-numeric series	Physics 2: Force, laws of motion, momentum, friction
3	Sun 4 Oct	Profit, loss & discount	Analogy	Physics 3: Work, energy, power + gravitation
4	Mon 5 Oct	Simple & compound interest	Classification (odd one out)	Physics 4: Pressure, fluids, properties of matter + heat
5	Tue 6 Oct	Number system	Coding-decoding	Physics 5: Waves & sound
6	Wed 7 Oct	LCM & HCF, simplification, surds	Blood relations	Physics 6: Light
7	Thu 8 Oct	Average + ages	Direction sense	Physics 7: Electricity, magnetism + inventions
8	Fri 9 Oct	Mixture & alligation	Order & ranking, dictionary order	PHYSICS REVISION: mixed set + mistake log
9	Sat 10 Oct	Time & work, pipes & cisterns	Mathematical operations	Chemistry 1: Matter, mixtures, atomic structure
10	Sun 11 Oct	Time, speed & distance	Venn diagrams	Chemistry 2: Periodic table & bonding
11	Mon 12 Oct	Trains, boats & streams	Syllogism	Chemistry 3: Chemical reactions
12	Tue 13 Oct	Mensuration 2D	Statement-conclusion / assumption	Chemistry 4: Acids, bases, salts
13	Wed 14 Oct	Mensuration 3D	Data sufficiency	Chemistry 5: Metals, non-metals, metallurgy
14	Thu 15 Oct	Algebra	Seating arrangement	Chemistry 6: Carbon compounds, polymers, fuels

Day	Date	Maths	Reasoning	Science
15	Fri 16 Oct	Geometry I: lines, triangles, polygons	Puzzles	Chemistry 7: Everyday chemistry
16	Sat 17 Oct	Geometry II: circles	Mirror & water images	CHEMISTRY REVISION: mixed set + mistake log
17	Sun 18 Oct	Trigonometry: ratios & identities	Paper folding/cutting, embedded figures	Biology 1: Cell, cell division, tissues
18	Mon 19 Oct	Heights & distances	Figure counting & completion	Biology 2: Classification, microbes, diseases
19	Tue 20 Oct	Statistics & data interpretation	Dice & cubes	Biology 3: Nutrition, digestion, vitamins
20	Wed 21 Oct	Calendar & clock	Missing number (matrix/figure)	Biology 4: Respiration, circulation, excretion
21	Thu 22 Oct	Mixed PYQ set: arithmetic	Word formation, jumbling, decision making	Biology 5: Nervous, endocrine, reproduction
22	Fri 23 Oct	Mixed PYQ set: mensuration, algebra, geometry	Mixed timed set	Biology 6: Plant biology
23	Sat 24 Oct	Full mock: maths section	Full mock: reasoning section	Biology 7: Genetics, evolution, ecology
24	Sun 25 Oct	Full CBT mock + analysis	(in mock)	BIOLOGY REVISION: mixed set + mistake log
25	Mon 26 Oct	Formula sheet + mistake log only	Mistake log only	Science traps + mistake log only. Sleep early.

Science split: Physics 7 days + 1 revision · Chemistry 7 + 1 · Biology 7 + 1 · final day mixed. Puzzles and seating arrangements come later because they take the longest to pay off.

SCIENCE — Day 1 · Physics 1: Units, Measurement & Motion

Reading time about 30 min. Learn the **tables** and the **traps**; examiners mostly test those. After each part, answer the Check questions before moving on.

Part A — Units & Measurement

A1. Fundamental vs derived quantities

A **fundamental** quantity stands on its own and isn't made from other quantities. A **derived** quantity is built from fundamental ones. For example, speed = length ÷ time. The SI system has exactly **7 base units**:

m metre Length	kg kilogram Mass	s second Time	A ampere Electric current	K kelvin Temperature	mol mole Amount of substance	cd candela Luminous intensity
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Remember

- **kg** is the only base unit with a prefix ('kilo') in its name.
- Temperature is in **kelvin (K)**, not °C. It's written 'K', not '°K'.
- **Candela** is luminous intensity. Lumen (flux) and lux (illuminance) are derived units, not base units.
- Radian (plane angle) and steradian (solid angle) used to be 'supplementary' units. They are now treated as derived (dimensionless).

A2. Derived units: the high-frequency list

Quantity	SI unit (symbol)	In base units / meaning
Force	newton (N)	$\text{kg}\cdot\text{m}/\text{s}^2$
Work, energy, heat	joule (J)	$\text{N}\cdot\text{m} = \text{kg}\cdot\text{m}^2/\text{s}^2$
Power	watt (W)	J/s
Pressure, stress	pascal (Pa)	N/m^2
Frequency	hertz (Hz)	1/s
Electric charge	coulomb (C)	$\text{A}\cdot\text{s}$
Potential difference, EMF	volt (V)	$\text{J}/\text{C} = \text{W}/\text{A}$
Resistance	ohm (Ω)	V/A
Conductance	siemens (S)	$1/\Omega$ (older name: mho)
Capacitance	farad (F)	C/V
Magnetic flux	weber (Wb)	$\text{V}\cdot\text{s}$
Magnetic field (flux density)	tesla (T)	Wb/m^2
Inductance	henry (H)	Wb/A
Luminous flux / illuminance	lumen (lm) / lux (lx)	$\text{lux} = \text{lm}/\text{m}^2$
Radioactivity	becquerel (Bq)	1 decay per second (older unit: curie)
Power of a lens	diopetre (D)	1/focal length in metres

A3. Special units and conversions: the most-tested trap area

Unit	What it measures	Value
Light year	DISTANCE (not time)	$9.46 \times 10^{15} \text{ m}$ (distance light travels in 1 year)
Parsec	Distance (astronomy)	$3.26 \text{ light years} \approx 3.08 \times 10^{16} \text{ m}$ — largest of the three
Astronomical unit (AU)	Distance	$1.496 \times 10^{11} \text{ m}$ (mean Sun–Earth distance)
Ångström (Å)	Very small length (wavelength, atoms)	10^{-10} m
Fermi (fm)	Nuclear size	10^{-15} m
Micron (μm)	Small length	10^{-6} m
Nautical mile	Distance at sea and in air	1852 m
Knot	SPEED of ships	1 nautical mile per hour
Fathom	Depth of water	6 feet $\approx 1.83 \text{ m}$
Horsepower (HP)	Power	1 HP = 746 W
Kilowatt-hour (kWh)	ENERGY (the electricity 'unit')	$3.6 \times 10^6 \text{ J}$
Calorie	Heat energy	1 cal $\approx 4.186 \text{ J}$ (often taken as 4.2 J)
Electron-volt (eV)	Energy of particles	$1.6 \times 10^{-19} \text{ J}$
Atmosphere (atm)	Pressure	$1.013 \times 10^5 \text{ Pa} = 76 \text{ cm (760 mm) of Hg}$
Bar / torr	Pressure	1 bar = 10^5 Pa ; 1 torr = 1 mm of Hg
Decibel (dB)	Sound intensity level (loudness)	Logarithmic scale

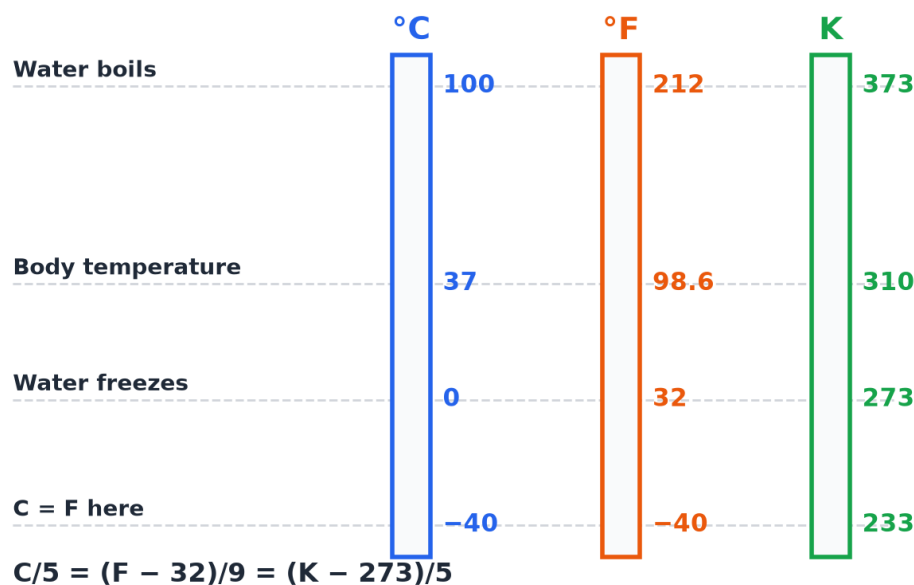
Traps (examiners love these)

- Light year → **distance**, not time. Knot → **speed**, not distance. kWh → **energy**, not power.
- Distance units in increasing order: **Å** < **μm** < **km** < **AU** < **light year** < **parsec**.
- 'Unit of electricity' on your bill = 1 kWh = 3.6 MJ.

Check A1-A3 (answer before reading on)

a) SI unit of luminous intensity? b) The 'unit' on an electricity bill is a unit of what? c) Which is larger: 1 parsec or 1 light year?

A4. Temperature scales



Method: put the known value into $C/5 = (F - 32)/9 = (K - 273)/5$. **Example:** 25 °C to °F: $25/5 = 5 \rightarrow F - 32 = 9 \times 5 = 45 \rightarrow F = 77$ °F. In kelvin: $K = 25 + 273 = 298$ K.

Facts: -40 °C = -40 °F (the only point where C = F). Absolute zero = 0 K = -273.15 °C. A **change** of 1 °C equals a change of 1 K, which equals a change of 1.8 °F.

A5. Measuring instruments

Instrument	Measures	Instrument	Measures
Barometer	Atmospheric pressure	Hygrometer	Relative humidity
Hydrometer	Relative density of liquids	Lactometer	Purity of milk
Manometer	Pressure of gases	Sphygmomanometer	Blood pressure
Ammeter	Current (connected in series)	Voltmeter	Potential difference (connected in parallel)
Galvanometer	Detects small currents	Anemometer	Wind speed
Altimeter	Altitude	Seismograph	Earthquake waves
Pyrometer	Very high temperatures	Clinical thermometer	Body temp: 35-42 °C (94-108 °F)
Odometer	Distance travelled by a vehicle	Tachometer	Rotation speed (rpm)
Vernier calliper	Least count 0.1 mm = 0.01 cm	Screw gauge	Least count 0.01 mm = 0.001 cm

Hydrometer vs hygrometer vs hydrophone

Hydrometer → density of liquids. **Hygrometer** → humidity ('gro' as in grow, which needs moisture). **Hydrophone** → sound underwater.

A6. Scalars vs vectors

A **scalar** has only magnitude (e.g. 5 kg). A **vector** has magnitude and direction (e.g. 5 N towards east).

Scalars	Vectors
Mass, distance, speed, time, work, energy, power, temperature, pressure, density, volume, charge, electric current	Displacement, velocity, acceleration, force, weight, momentum, impulse, torque, electric field, magnetic field

Traps

- **Electric current** has a direction but is a **scalar**, because currents don't add like vectors.
- **Pressure** is a scalar, but force is a vector. **Weight** is a vector (a force), but mass is a scalar.

A7. Dimensions (M = mass, L = length, T = time)

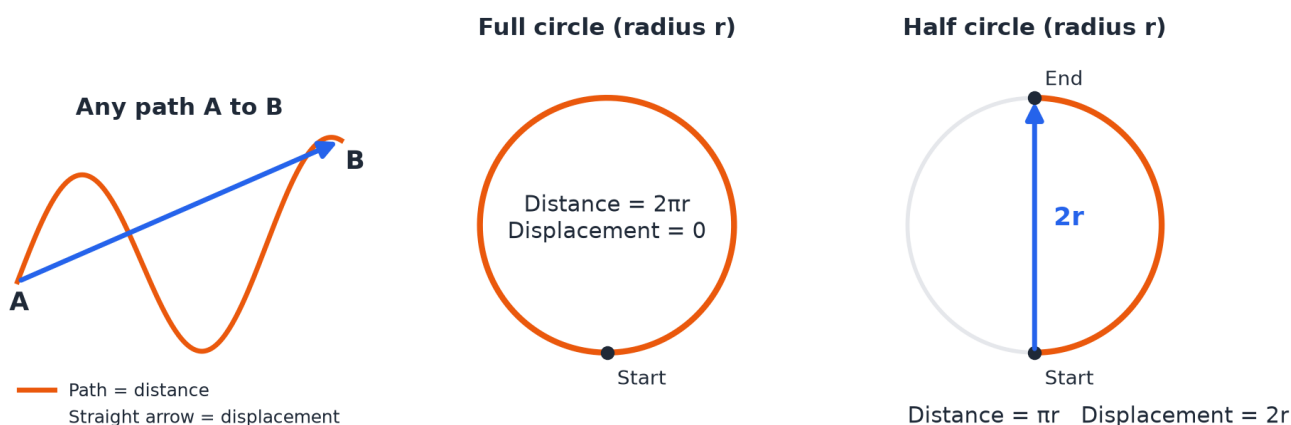
Quantity	Dimension	Quantity	Dimension
Velocity	$[L T^{-1}]$	Acceleration	$[L T^{-2}]$
Force	$[M L T^{-2}]$	Momentum, impulse	$[M L T^{-1}]$
Work, energy, torque	$[M L^2 T^{-2}]$	Power	$[M L^2 T^{-3}]$
Pressure, stress, Young's modulus	$[M L^{-1} T^{-2}]$	Planck's constant, angular momentum	$[M L^2 T^{-1}]$
Gravitational constant G	$[M^{-1} L^3 T^{-2}]$	Frequency	$[T^{-1}]$

Same-dimension pairs (frequent MCQ): work-energy-torque · pressure-stress-Young's modulus · impulse-momentum · Planck's constant-angular momentum. **Dimensionless:** strain, angle, refractive index, relative density.

Part B — Motion (kinematics)

B1. Distance vs displacement

Distance is the full length of the path you actually travelled (scalar). **Displacement** is the straight line from start to end, with direction (vector).



Rule	Why it matters
Displacement $\leq$ distance (always)	They are equal only for straight-line motion in one direction.
Displacement can be zero or negative; distance cannot	Any round trip or full circle has displacement 0.
Full circle: distance $2\pi r$, displacement 0	Half circle: distance πr , displacement $2r$ (the diameter).

B2. Speed vs velocity, and the average-speed trap

Speed = distance $\div$ time (scalar). Velocity = displacement $\div$ time (vector). Average speed = **total distance $\div$ total time**. It is **not** the simple average of the speeds.

Worked example

You go 60 km at 60 km/h and return at 40 km/h. Average speed?

Time going = 1 h, time returning = 1.5 h. Total distance 120 km $\div$ 2.5 h = **48 km/h** (not 50).

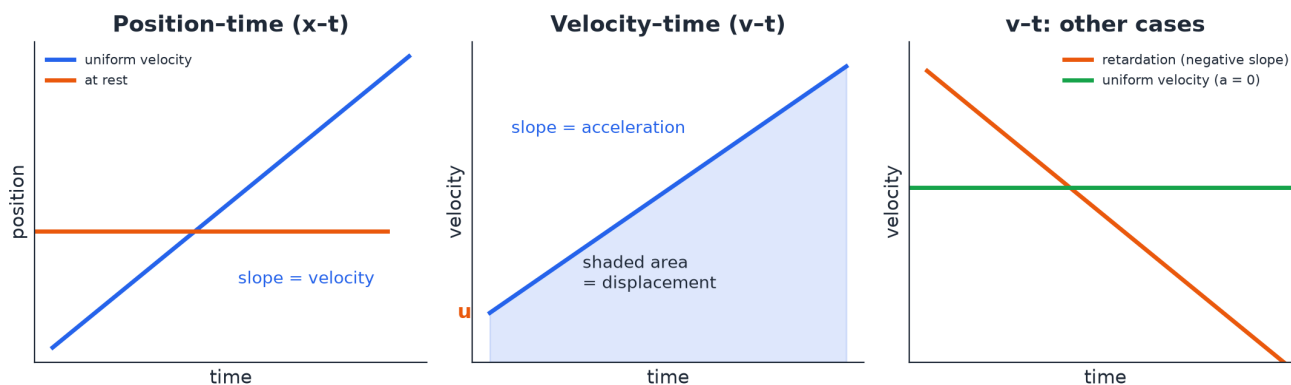
Shortcut for equal distances: **$2xy/(x + y)$** = $2 \times 60 \times 40 / 100 = 48$ km/h. Average velocity for the round trip = 0.

Converting units: km/h $\rightarrow$ m/s: multiply by **5/18**. m/s $\rightarrow$ km/h: multiply by **18/5**. Memorise: 18 km/h = 5 m/s $\cdot$ 36 = 10 $\cdot$ 54 = 15 $\cdot$ 72 = 20 $\cdot$ 90 = 25 $\cdot$ 108 = 30.

B3. Acceleration

Acceleration = change in velocity $\div$ time (m/s^2). Negative acceleration is called **retardation** (deceleration). Velocity changes if **either** speed or direction changes. So in **uniform circular motion** the speed is constant but the velocity keeps changing, which means it **is** accelerated motion (acceleration points towards the centre).

B4. Motion graphs: slope and area



Graph	Slope gives	Area under it gives	Straight horizontal line means
Position-time (x-t)	Velocity	Nothing useful	Body at rest
Velocity-time (v-t)	Acceleration	Displacement	Uniform velocity ($a = 0$)
Acceleration-time (a-t)	–	Change in velocity	Uniform acceleration

B5. Equations of motion (for uniform acceleration only)

#	Equation	Use it when the question gives no...
1	$v = u + at$	distance (s)
2	$s = ut + \frac{1}{2}at^2$	final velocity (v)
3	$v^2 = u^2 + 2as$	time (t)
4	Distance in the nth second: $s_n = u + a(2n - 1)/2$	- (used for 'distance in the 5th second' questions)

u = initial velocity · v = final velocity · a = acceleration · t = time · s = displacement. 'Starts from rest' means $u = 0$. 'Comes to rest' or 'stops' means $v = 0$.

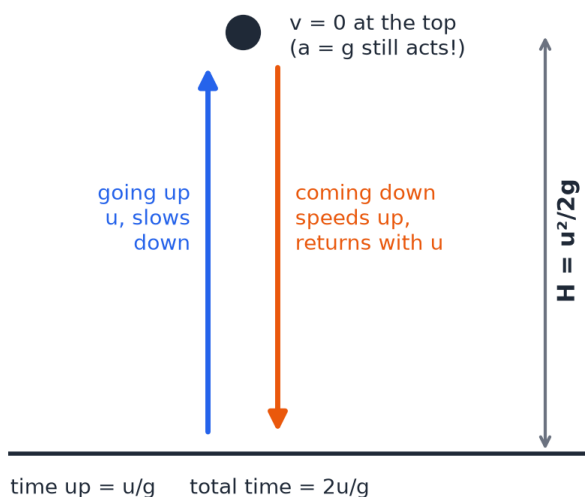
Worked example

A car starts from rest and reaches 20 m/s in 5 s. Find a and s .

$a = (v - u)/t = (20 - 0)/5 = 4 \text{ m/s}^2$. $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 4 \times 25 = 50 \text{ m}$.

Check with $v^2 = 2as$: $400 = 2 \times 4 \times s \rightarrow s = 50 \text{ m}$ ✓

B6. Free fall and vertical throw ($g \approx 9.8 \text{ m/s}^2$, use 10 for quick sums)



Situation	Formulas
Dropped from rest	$v = gt$ · $h = \frac{1}{2}gt^2$ · $v^2 = 2gh$
Thrown up with speed u	Max height $H = u^2/2g$ · time to the top = u/g · total time in the air = $2u/g$
At the highest point	Velocity = 0, but acceleration = g (downwards) — not zero
In a vacuum	A feather and a coin fall together: all bodies have the same acceleration (Galileo)

Worked example

A ball is thrown up at 20 m/s ($g = 10$). $H = 400/20 = 20 \text{ m}$. Time up = $20/10 = 2 \text{ s}$. Total time = **4 s**. It lands at 20 m/s.

Day 1 — top 10 traps (re-read these before bed)

1. Light year = distance. 2. Knot = speed. 3. kWh = energy = $3.6 \times 10^6 \text{ J}$. 4. 1 HP = 746 W.
5. Electric current and pressure are **scalars**. 6. $-40^\circ \text{C} = -40^\circ \text{F}$.
7. Full circle: displacement = 0. 8. Average speed for equal distances = $2xy/(x + y)$, not $(x + y)/2$.
9. Uniform circular motion is accelerated. 10. At the top of a throw, $v = 0$ but $a = g$.

Day 1 — Science practice set (20 Qs · target 6 minutes)

Write all your answers first, then check against the key on the next page. Send me your score and the question numbers you got wrong.

#	Question
1	SI unit of luminous intensity is: (a) lumen (b) lux (c) candela (d) watt
2	Light year is a unit of: (a) time (b) distance (c) speed (d) intensity of light
3	1 kWh equals: (a) 3.6×10^3 J (b) 3.6×10^5 J (c) 3.6×10^6 J (d) 36×10^6 J
4	Which is a vector? (a) speed (b) work (c) momentum (d) energy
5	The relative density of a liquid is measured by a: (a) hygrometer (b) hydrometer (c) lactometer (d) manometer
6	At what temperature are the Celsius and Fahrenheit readings equal? (a) 0 (b) -40 (c) 40 (d) -273
7	A body completes one full circle of radius 7 m. Its displacement is: (a) 44 m (b) 14 m (c) 0 (d) 22 m
8	A body goes half way round a circle of radius 7 m ($\pi = 22/7$). Its distance and displacement are: (a) 22 m, 14 m (b) 14 m, 22 m (c) 44 m, 0 (d) 22 m, 0
9	The slope of a velocity-time graph gives: (a) displacement (b) speed (c) acceleration (d) jerk
10	The area under a velocity-time graph gives: (a) acceleration (b) displacement (c) force (d) momentum
11	A car accelerates from rest at 2 m/s^2 for 10 s. The distance covered is: (a) 20 m (b) 50 m (c) 100 m (d) 200 m
12	54 km/h in m/s is: (a) 12 (b) 15 (c) 18 (d) 20
13	A ball is thrown up at 30 m/s ($g = 10 \text{ m/s}^2$). Its maximum height is: (a) 30 m (b) 45 m (c) 90 m (d) 60 m
14	Which quantity has the same dimensions as work? (a) power (b) force (c) torque (d) momentum
15	A person goes at 60 km/h and returns the same distance at 40 km/h. Their average speed is: (a) 50 (b) 48 (c) 45 (d) 52 km/h
16	Pascal is the unit of: (a) force (b) pressure (c) energy (d) power
17	The least count of a screw gauge is: (a) 0.1 mm (b) 0.01 mm (c) 0.1 cm (d) 1 mm
18	Which is NOT a unit of distance? (a) parsec (b) AU (c) knot (d) light year
19	The SI unit of the power of a lens is: (a) watt (b) dioptre (c) metre (d) joule
20	In uniform circular motion, which stays constant? (a) velocity (b) acceleration vector (c) speed (d) displacement

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Day 1 — Answer key with reasons

#	Ans	Why
1	c	Candela is the base unit. Lumen = luminous flux, lux = illuminance.
2	b	Distance light travels in one year: 9.46×10^{15} m.
3	c	$1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$.
4	c	Momentum = mass $\times$ velocity, so it has direction.
5	b	Hygrometer = humidity, lactometer = milk, manometer = gas pressure.
6	b	Put $C = F$ into $C/5 = (F - 32)/9$: $9C = 5C - 160 \rightarrow C = -40$.
7	c	Start point = end point.
8	a	Distance = $\pi r = 22/7 \times 7 = 22$ m. Displacement = diameter = 14 m.
9	c	Change in velocity $\div$ time.
10	b	Area under v-t = displacement.
11	c	$s = \frac{1}{2} \times 2 \times 10^2 = 100$ m.
12	b	$54 \times 5/18 = 15$.
13	b	$H = u^2/2g = 900/20 = 45$ m.
14	c	Torque = force $\times$ distance $\rightarrow [M L^2 T^{-2}]$, the same as work.
15	b	$2 \times 60 \times 40 / (60 + 40) = 48$.
16	b	$\text{Pa} = \text{N/m}^2$.
17	b	Screw gauge 0.01 mm; vernier 0.1 mm.
18	c	Knot = speed (nautical mile per hour).
19	b	Power = $1/f$ (in metres) $\rightarrow$ dioptre.
20	c	The direction changes all the time, so only the speed is constant.

Check A1-A3 answers

a) candela b) energy (kWh) c) 1 parsec (= 3.26 light years)

SCIENCE — Days 2 to 25: topic map

Full illustrated notes for each day will be prepared in the same format as Day 1. Ask for 'Science Day N' the evening before. Use this map to see what's coming.

Day	Subject	Topic	Must-cover points
2	Physics	Force & laws of motion	Newton's 3 laws & examples, inertia types, momentum & conservation (recoil of a gun), impulse, friction (static > kinetic > rolling), centripetal force, banking of roads
3	Physics	Work, energy, power + gravitation	$W = Fs \cos\theta$, $KE = \frac{1}{2}mv^2$, $PE = mgh$, energy conversions in devices, power; g , variation of g (poles > equator, zero at the centre), weight vs mass, escape velocity 11.2 km/s, satellites, Kepler's laws
4	Physics	Pressure, fluids, matter, heat	Pascal's law (hydraulic lift), Archimedes, buoyancy & floating, density of ice/water, surface tension, capillarity, viscosity, Bernoulli; specific heat, latent heat, expansion, conduction/convection/radiation, anomalous expansion of water at 4 °C
5	Physics	Waves & sound	Longitudinal vs transverse, speed in solid > liquid > gas, echo (17 m rule), infrasonic below 20 Hz / ultrasonic above 20 kHz, SONAR, Doppler effect, pitch-frequency, loudness-amplitude, decibel
6	Physics	Light	Laws of reflection, plane/concave/convex mirrors and their uses, refraction & refractive index, total internal reflection (optical fibre, mirage, diamond), lenses, eye defects & correction, dispersion (VIBGYOR), scattering (blue sky, red sunset), rainbow
7	Physics	Electricity & magnetism	Ohm's law, resistances in series/parallel, resistivity, electric power & kWh bill sums, heating effect, fuse, domestic wiring, magnetic field, electromagnet, motor vs generator, EMI, AC vs DC, transformer; scientists & inventions
8	Physics	REVISION	50 mixed questions + redo every Physics mistake from Days 1-7
9	Chemistry	Matter & atomic structure	States of matter, sublimation, mixtures vs compounds, separation methods (chromatography, distillation, centrifugation), Dalton → Thomson → Rutherford → Bohr, electron/proton/neutron discoverers, isotopes/isobars/isotones, valency
10	Chemistry	Periodic table & bonding	Mendeleev vs modern table, groups & periods, trends (atomic size, metallic character, electronegativity), noble gases, ionic vs covalent bonds, properties of ionic compounds
11	Chemistry	Chemical reactions	Combination, decomposition, displacement, double displacement, redox (OIL RIG), exo/endothermic, corrosion & rancidity, balancing, catalysts
12	Chemistry	Acids, bases, salts	Properties, indicators (litmus, phenolphthalein, methyl orange), pH scale & everyday pH values, naturally occurring acids (lemon-citric, vinegar-acetic, ant sting-formic), common salts: baking soda, washing soda, bleaching powder, plaster of Paris
13	Chemistry	Metals & non-metals	Physical/chemical properties, reactivity series, ores of Al/Fe/Cu/Zn, metallurgy steps, alloys & their composition (brass, bronze, steel, solder, amalgam), exceptions (Hg liquid, Na soft, graphite conducts)
14	Chemistry	Carbon & its compounds	Catenation, allotropes (diamond, graphite, fullerene), hydrocarbons (alkane/alkene/alkyne), functional groups, ethanol & ethanoic acid, soaps vs detergents, polymers (PVC, nylon, Teflon, Bakelite), fuels & calorific value, LPG/CNG composition

Day	Subject	Topic	Must-cover points
15	Chemistry	Everyday chemistry	Common names & formulas (dry ice, laughing gas, heavy water, quicklime, slaked lime, etc.), gases & their uses, fertilisers, glass & cement, hard water, radioactivity basics
16	Chemistry	REVISION	50 mixed questions + redo every Chemistry mistake
17	Biology	Cell	Cell theory, prokaryotic vs eukaryotic, plant vs animal cell, organelles & functions (mitochondria, ribosome, lysosome 'suicide bags', Golgi, ER, chloroplast), mitosis vs meiosis, plant & animal tissues
18	Biology	Classification, microbes, diseases	5-kingdom (Whittaker), binomial nomenclature, scientific names of common organisms, bacteria/virus/fungi/protozoa, diseases by pathogen, vaccines & discoverers
19	Biology	Nutrition & digestion	Nutrients, vitamins (fat- vs water-soluble, sources, deficiency diseases), minerals (Fe, I, Ca), digestive organs & enzymes, liver/bile, absorption in the small intestine
20	Biology	Respiration, circulation, excretion	Aerobic vs anaerobic respiration, lungs & alveoli, heart chambers & blood flow, arteries vs veins, blood components, blood groups (O universal donor, AB universal recipient), kidney & nephron
21	Biology	Control & reproduction	Neuron, brain parts & functions, reflex action, endocrine glands & hormones (insulin, thyroxine, adrenaline, growth hormone), human reproduction basics, sense organs (eye, ear)
22	Biology	Plant biology	Photosynthesis equation & site, stomata, transpiration, xylem vs phloem, plant hormones (auxin, gibberellin, cytokinin, ABA, ethylene), pollination & fertilisation, tropisms
23	Biology	Genetics, evolution, ecology	Mendel's laws, dominant/recessive, sex determination (XX/XY), DNA, evolution (Darwin, Lamarck), food chains & 10% law, ecosystems, ozone, greenhouse effect
24	Biology	REVISION	50 mixed questions + redo every Biology mistake
25	All	Final day	Top traps from all three subjects + your full mistake log. No new topics.

MATHS — Day-wise topics, question patterns & formulas

For each day: read the formulas (5 min), then practise each listed pattern from your book. Aim for 2–3 questions per pattern, timed.

Day 1 · Percentage

Question patterns to practise	Formulas
% of a number % increase / decrease - Successive % change - A is r% more than B → B is ?% less than A - Price–consumption–expenditure - Population growth / depreciation - Elections & votes - Marks: pass/fail	- Successive changes a% and b%: $a + b + \frac{ab}{100}$ (use –ve for a decrease) - A is r% more than B → B is $\frac{r}{(100 + r)} \times 100\%$ less than A - A is r% less than B → B is $\frac{r}{(100 - r)} \times 100\%$ more than A - Price up r% → reduce consumption by $\frac{r}{(100 + r)} \times 100\%$ to keep expenditure the same - Population after n years: $P(1 + \frac{r}{100})^n$ · depreciation: $P(1 - \frac{r}{100})^n$ - Fraction ↔ %: $\frac{1}{2} = 50$ · $\frac{1}{3} = 33\frac{1}{3}$ · $\frac{1}{4} = 25$ · $\frac{1}{5} = 20$ · $\frac{1}{6} = 16\frac{2}{3}$ · $\frac{1}{7} \approx 14.28$ · $\frac{1}{8} = 12.5$ · $\frac{1}{9} \approx 11.11$ · $\frac{1}{11} \approx 9.09$ · $\frac{1}{12} = 8\frac{1}{3}$ · $\frac{1}{16} = 6.25$

Day 2 · Ratio & proportion, partnership

Question patterns to practise	Formulas
- Combining ratios (A:B, B:C → A:B:C) - Dividing an amount in a ratio - Number to add or subtract to reach a ratio - Mean / third / fourth proportional - Coins & notes - Income–expenditure–savings - Partnership profit sharing	$a:b = c:d \rightarrow ad = bc$ - Mean proportional of a, b = $\sqrt{ab}$ · third proportional = $\frac{b^2}{a}$ · fourth proportional of a, b, c = $\frac{bc}{a}$ - Duplicate ratio $a^2:b^2$ · sub-duplicate $\sqrt{a}:\sqrt{b}$ · triplicate $a^3:b^3$ - Partnership: profit share $\propto \text{capital} \times \text{time}$

Day 3 · Profit, loss & discount

Question patterns to practise	Formulas
- Basic profit / loss % - Finding CP or SP - Marked price & discount - Successive discounts - Dishonest dealer (false weights) - Two items sold at the same SP, one at a gain and one at a loss - Buy x get y free	- Profit% = $\frac{(SP - CP)}{CP} \times 100$ — always on CP - SP = $CP \times \frac{(100 + P\%)}{100}$ · SP = $CP \times \frac{(100 - L\%)}{100}$ - Discount is on MP: SP = $MP \times \frac{(100 - d)}{100}$ - Successive discounts a%, b% = $a + b - \frac{ab}{100}$ - False weight: gain% = $\frac{\text{error}}{(\text{true value} - \text{error})} \times 100$ - Same SP, one at x% gain and one at x% loss → always a loss of $\frac{x^2}{100}\%$

Day 4 · Simple & compound interest

Question patterns to practise	Formulas
- SI basic - Money doubles / triples - CI annually & half-yearly - CI – SI difference - Instalments - Population-style growth	- SI = $\frac{PRT}{100}$ - Amount (CI) = $P(1 + \frac{R}{100})^n$ - Half-yearly: rate $\frac{R}{2}$, time $2n$ · quarterly: $\frac{R}{4}$, $4n$ - CI – SI for 2 years = $\frac{P(R/100)^2}{2}$ · for 3 years = $\frac{P(R/100)^2(3 + \frac{R}{100})}{2}$ - SI: sum doubles in T years → $R = \frac{100}{T}$ - CI: sum becomes x times in n years → x^k times in kn years

Day 5 · Number system

Question patterns to practise	Formulas
- Divisibility tests - Unit digit of powers - Remainders - Number of factors - Sum of series - Find the missing digit	- Divisibility: $3/9 \rightarrow$ digit sum $\cdot 4 \rightarrow$ last 2 digits $\cdot 8 \rightarrow$ last 3 digits $\cdot 11 \rightarrow$ (odd-place sum – even-place sum) = 0 or a multiple of 11 - Unit digit cycles: 2, 3, 7, 8 $\rightarrow$ cycle of 4 $\cdot 4, 9 \rightarrow$ cycle of 2 $\cdot 0, 1, 5, 6 \rightarrow$ unchanged - Sum of first n natural numbers = $\frac{n(n+1)}{2}$ $\cdot$ squares = $\frac{n(n+1)(2n+1)}{6}$ $\cdot$ cubes = $\left[\frac{n(n+1)}{2}\right]^2$ $\cdot$ first n odd numbers = n^2 $N = a^p \times b^q \rightarrow$ number of factors = $(p+1)(q+1)$ - Dividend = divisor $\times$ quotient + remainder

Day 6 · LCM & HCF, simplification, surds

Question patterns to practise	Formulas
- Largest number dividing x, y, z leaving remainder r - Smallest number leaving remainder r - Bells / lights together - HCF & LCM of fractions - BODMAS simplification - Square roots, surds, approximation	LCM $\times$ HCF = product of the two numbers (two numbers only) - HCF of fractions = HCF(numerators)/LCM(denominators) $\cdot$ LCM of fractions = LCM(numerators)/HCF(denominators) - Largest number dividing a, b, c leaving the same remainder = HCF of $(a-b), (b-c), (c-a)$ - Smallest number that leaves remainder r when divided by x, y, z = LCM(x, y, z) + r - Order: Brackets $\rightarrow$ Of $\rightarrow$ Division $\rightarrow$ Multiplication $\rightarrow$ Addition $\rightarrow$ Subtraction

Day 7 · Average + ages

Question patterns to practise	Formulas
- Simple average - Average of consecutive numbers - One member replaced / added / removed - Wrong value recorded - Cricket batting average - Present / past / future ages in a ratio	- Average = sum / count $\cdot$ sum = average $\times$ count - Average of first n natural numbers = $\frac{(n+1)}{2}$ $\cdot$ average of consecutive numbers = (first + last)/2 - Average of n items rises by x when one is replaced $\rightarrow$ new – old = $n \times x$ - Ages: every person ages by the same number of years — use a ratio-multiplier k

Day 8 · Mixture & alligation

Question patterns to practise	Formulas
- Mixing two prices - Milk–water ratio - Repeated replacement - Mixing two solutions of different %	- Alligation: cheaper : dearer = $(d - m) : (m - c)$ (m = mean price) - After n replacements of x litres from V litres: pure left = $V(1 - \frac{x}{V})^n$

Day 9 · Time & work, pipes & cisterns

Question patterns to practise	Formulas
- A and B together - Efficiency-based - Work and wages - MDH (men–days–hours) - Someone leaves part way - Pipe and leak	- A alone a days, B alone b days $\rightarrow$ together $\frac{ab}{(a+b)}$ days - LCM method: total work = LCM of days; efficiency = work/day $M_1D_1H_1/W_1 = M_2D_2H_2/W_2$ - Wages are shared in the ratio of work done (efficiency) - Pipes: filling pipe +, emptying pipe / leak –

Day 10 · Time, speed & distance

Question patterns to practise	Formulas
- Basic $S = D/T$ - Average speed - Relative speed - Late / early arrival - Ratio of speeds - Races	$S = D/T$ · km/h $\times 5/18 =$ m/s - Average speed for equal distances = $2xy/(x + y)$ - Same direction: relative speed = difference · opposite directions: sum - Same distance: speed ratio $a : b \rightarrow$ time ratio $b : a$

Day 11 · Trains, boats & streams

Question patterns to practise	Formulas
- Train crosses a pole / man - Train crosses a platform / bridge - Two trains crossing - Upstream / downstream - Speed of the boat & of the stream	- Pole: $t = L/S$ · platform: $t = (L_1 + L_2)/S$ - Two trains: $t = (L_1 + L_2)/(S_1 \pm S_2)$ (+ for opposite directions) - Downstream = $u + v$ · upstream = $u - v$ - Boat speed $u = (D + U)/2$ · stream speed $v = (D - U)/2$

Day 12 · Mensuration 2D

Question patterns to practise	Formulas
- Rectangle / square - Triangles (incl. equilateral, Heron) - Circle, sector, arc - Trapezium, rhombus, parallelogram - Path / ring area % change in area	- Rectangle lb, perimeter $2(l + b)$, diagonal $\sqrt{l^2 + b^2}$ · square a^2, diagonal $a\sqrt{2}$ - Triangle $\frac{1}{2}bh$ · Heron $\sqrt{s(s - a)(s - b)(s - c)}$, $s = (a + b + c)/2$ - Equilateral: area $(\sqrt{3}/4)a^2$, height $(\sqrt{3}/2)a$ - Circle πr^2, circumference $2\pi r$ · semicircle perimeter $\pi r + 2r$ - Sector $(\theta/360)\pi r^2$ · arc $(\theta/360)2\pi r$ · ring $\pi(R^2 - r^2)$ - Trapezium $\frac{1}{2}(a + b)h$ · rhombus $\frac{1}{2}d_1d_2$ · parallelogram bh

Day 13 · Mensuration 3D

Question patterns to practise	Formulas
- Cube / cuboid - Cylinder - Cone - Sphere / hemisphere - Melting & recasting - Frustum, prism, pyramid	- Cube a^3, TSA $6a^2$, diagonal $a\sqrt{3}$ · cuboid lbh, TSA $2(lb + bh + hl)$, diagonal $\sqrt{l^2 + b^2 + h^2}$ - Cylinder $\pi r^2 h$, CSA $2\pi rh$, TSA $2\pi r(r + h)$ - Cone $\frac{1}{2}\pi r^2 h$, slant $l = \sqrt{r^2 + h^2}$, CSA πrl, TSA $\pi r(r + l)$ - Sphere $(4/3)\pi r^3$, SA $4\pi r^2$ · hemisphere $(2/3)\pi r^3$, CSA $2\pi r^2$, TSA $3\pi r^2$ - Frustum volume $(\pi h/3)(R^2 + r^2 + Rr)$ · prism = base area $\times h$ · pyramid = $\frac{1}{3} \times$ base area $\times h$ - Recasting: volume stays the same

Day 14 · Algebra

Question patterns to practise	Formulas
- Identities $x + 1/x$ type $a + b + c = 0$ type - Linear equations - Quadratic roots	$(a \pm b)^2 = a^2 \pm 2ab + b^2$ · $a^2 - b^2 = (a + b)(a - b)$ $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$ · $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ · $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \rightarrow$ if $a + b + c = 0$, $a^3 + b^3 + c^3 = 3abc$ $x + 1/x = k \rightarrow x^2 + 1/x^2 = k^2 - 2$ · $x^3 + 1/x^3 = k^3 - 3k$ $ax^2 + bx + c = 0$: sum of roots $-b/a$, product c/a, discriminant $b^2 - 4ac$

Day 15 · Geometry I: lines, triangles, polygons

Question patterns to practise	Formulas
- Angles on parallel lines - Triangle angle sums - Centroid / incentre / circumcentre / orthocentre - Similar triangles - Polygons	- Triangle angle sum 180° · exterior angle = sum of the two opposite interior angles - Centroid divides each median in 2 : 1 · incentre: $\angle BIC = 90^\circ + A/2$ - Similar triangles: area ratio = (side ratio)² - Equilateral: inradius $a/(2\sqrt{3})$, circumradius $a/\sqrt{3}$ - n-gon: interior-angle sum $(n - 2) \times 180^\circ$, each exterior angle (regular) $360^\circ/n$, diagonals $n(n - 3)/2$

Day 16 · Geometry II: circles

Question patterns to practise	Formulas
- Angles in a circle - Chords - Tangents - Cyclic quadrilaterals - Intersecting chords / secants	- Angle at the centre = 2 × angle at the circumference · angle in a semicircle = 90° - Tangent ⊥ radius · two tangents from an external point are equal - Cyclic quadrilateral: opposite angles sum to 180° - Chord at distance d: length $2\sqrt{(r^2 - d^2)}$ · tangent from distance d: $\sqrt{(d^2 - r^2)}$ - Chords: $PA \cdot PB = PC \cdot PD$ · tangent-secant: $PT^2 = PA \cdot PB$

Day 17 · Trigonometry: ratios & identities

Question patterns to practise	Formulas
- Standard values - Identities - Complementary angles - Max / min values - Simplification	$\sin^2\theta + \cos^2\theta = 1$ · $1 + \tan^2\theta = \sec^2\theta$ · $1 + \cot^2\theta = \operatorname{cosec}^2\theta$ - Values ($0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$): $\sin = 0, \frac{1}{2}, 1/\sqrt{2}, \sqrt{3}/2, 1$ · $\cos =$ reverse order · $\tan = 0, 1/\sqrt{3}, 1, \sqrt{3}, \text{undefined}$ $\sin(90^\circ - \theta) = \cos\theta$ · $\tan(90^\circ - \theta) = \cot\theta$ · if $A + B = 90^\circ$, $\tan A \cdot \tan B = 1$ - Max of $a \sin\theta + b \cos\theta = \sqrt{(a^2 + b^2)}$ · $\sin 2\theta = 2\sin\theta \cos\theta$ · $\cos 2\theta = \cos^2\theta - \sin^2\theta$

Day 18 · Heights & distances

Question patterns to practise	Formulas
- Angle of elevation - Angle of depression - Two angles from two points - Shadow problems	$\tan\theta = \text{height/base}$ · angle of elevation = angle of depression (alternate angles) 30°: height : base = $1 : \sqrt{3}$ · 45°: $1 : 1$ · 60°: $\sqrt{3} : 1$ π radians = 180°

Day 19 · Statistics & data interpretation

Question patterns to practise	Formulas
- Mean / median / mode - Range - Pie chart - Bar / line / table DI	- Median: middle value after sorting (n even → average of the two middle values) - Empirical relation: Mode = 3 Median – 2 Mean - Pie chart: $100\% = 360^\circ \rightarrow 1\% = 3.6^\circ$

Day 20 · Calendar & clock

Question patterns to practise	Formulas
• Day of the week for a date • Leap years • Angle between the hands • Hands coinciding / opposite / at right angles	• Odd days: normal year 1, leap year 2 · 100 years = 5, 200 = 3, 300 = 1, 400 = 0 • Leap year: divisible by 4; a century year must be divisible by 400 • Angle between the hands = $30H - 5.5M$ • Per day: hands coincide 22 times, are opposite 22 times, and at right angles 44 times

Day 21-25 · Mixed PYQ sets & full mocks

Question patterns to practise	Formulas
• Mixed sets by area • Full timed mocks • Mistake-log redo	• No new formulas — revise the formula boxes above and your mistake log

REASONING — Day-wise topics & patterns

Day	Topic	Patterns to practise
1	Number series	Constant difference · difference of differences · ratio ($\times$, $\div$) · squares/cubes $\pm n$ · primes · alternating (two series mixed) · $\times n + k$ (e.g. $\times 1 + 1$, $\times 2 + 2$) · factorial-type
2	Alphabet & alpha-numeric series	Letter positions (A = 1 ... Z = 26, reverse Z = 1) · EJOTY (5, 10, 15, 20, 25) · skip patterns · letter-number-symbol sequences · 'which letter is 7th to the right of ...'
3	Analogy	Number analogy (square, cube, $\pm$) · letter analogy (position shift) · word analogy (worker-tool, product-raw material, country-capital, unit-quantity)
4	Classification (odd one out)	Number (prime, square, even/odd) · letter-group (vowel, position gaps) · word (category membership)
5	Coding-decoding	Letter shift ($+n/-n$) · reverse alphabet (opposite pairs: A-Z, B-Y...) · number coding (position sum) · word-code matching · substitution coding
6	Blood relations	Direct statements · coded relations (A + B means ...) · family tree / generation diagram · gender identification
7	Direction sense	Final direction · shortest distance (Pythagoras: 3-4-5, 6-8-10) · shadow problems (sunrise in the east $\rightarrow$ shadow to the west) · turning left/right by degrees
8	Order & ranking, dictionary order	Rank from top/bottom (total = top + bottom - 1) · interchange positions · comparison chains (A > B > C) · arranging words in dictionary order
9	Mathematical operations	Symbol substitution (+ means $\times$) · interchange of signs · BODMAS after substitution · balancing equations
10	Venn diagrams	Choosing the diagram that represents a relationship (e.g. doctors-men-fathers) · reading numbers from overlapping regions
11	Syllogism	All/Some/No statements · Venn-based method · 'either-or' cases · possibility conclusions
12	Statement-conclusion / assumption	Implicit assumptions · direct conclusions only (no outside knowledge) · course of action · cause-effect
13	Data sufficiency	Statement I alone / II alone / both / neither · apply to ranking, direction, blood relation, age
14	Seating arrangement	Linear (one row, two rows facing) · circular (facing the centre / outside) · fix definite clues first, then the conditional ones
15	Puzzles	Floor puzzles · box/stack puzzles · scheduling (days/months) · comparison puzzles · grid method
16	Mirror & water images	Mirror: left $\leftrightarrow$ right · water: top $\leftrightarrow$ bottom · clock mirror image (11:60 - time)
17	Paper folding/cutting, embedded figures	Unfold step by step (holes are mirrored across the fold) · find the hidden figure
18	Figure counting & completion	Counting triangles/squares systematically · completing a pattern · series of figures
19	Dice & cubes	Opposite faces from 2-3 dice views (common face rule) · standard dice (opposites sum to 7) · painted cube cut into n^3 : 3 faces painted = 8, 2 faces = $12(n - 2)$, 1 face = $6(n - 2)^2$, 0 = $(n - 2)^3$
20	Missing number (matrix/figure)	Row/column sum or product · centre = f(corners) · square/cube relations

Day	Topic	Patterns to practise
21	Word formation, jumbling, decision making	Make words from the letters of a given word · logical order of words · arranging events in sequence
22-25	Mixed timed sets & mocks	35-40 sec per question · skip anything over 1 min on the first pass

Mistake log (print or copy this format)

Mistake types: **C** = concept · **F** = formula · **Ca** = calculation · **I** = interpretation (misread) · **Cr** = careless · **R** = recall / memory. If the same type appears 3 times, write a rule for it and drill it.

[illegible]